



WILL JONES

SENIOR SOFTWARE ENGINEER

Will is a Senior Software Engineer specialising in cloud-native geospatial solutions, with more than eight years of experience delivering high-impact spatial platforms for clients across climate tech, environmental science, biodiversity, conservation, and land administration. He brings full-stack depth to every engagement, from designing scalable Azure infrastructure and automated data pipelines through to polished, map-driven frontends, with core expertise in Python, JavaScript/TypeScript, SQL, and open-source geospatial toolchains.

FEATURED PROJECTS

SPATIAL DATA FRAMEWORK FOR COASTAL AND OCEAN REPORTING

THE UNIVERSITY OF NEW SOUTH WALES

- Design and deliver a cloud-native spatial data platform for UNSW's Centre for Sustainable Development Reform, bringing together environmental datasets, processing pipelines, and reporting tools in a single transparent system.
- Build on open-source technologies to minimise vendor lock-in, ensuring UNSW and its partners can operate and extend the system independently.
- Support sustained capability through structured workshops and documentation designed to outlast the project delivery period.

LAND DEGRADATION NEUTRALITY FOR SIDS

CONSERVATION INTERNATIONAL

- Collaborated with Conservation International and the Pacific Community to deliver high-resolution land cover intelligence for Small Island Developing States.
- Design and deliver a 30-metre resolution annual land use and land cover dataset spanning 2000–2024, aligned with UNCCD reporting.
- Build an open-science processing pipeline using Landsat GeoMedian composites, machine learning classification, and cloud-native publication via Google Earth Engine, Zenodo, and Trends.Earth.
- Address a critical monitoring gap for SIDS, where existing global datasets are too coarse to capture the fine-grained land dynamics of small island landscapes.

UNESCO WORLD HERITAGE ONLINE MAP PLATFORM

- Led the creation of a global spatial monitoring system covering more than 10,000 World Heritage sites, managing the full project lifecycle as lead client liaison and team manager.
- Cleaned and validated UNESCO spatial data to authoritative standards and automated daily threat monitoring, including wildfire and deforestation detection, feeding a public dashboard and map.
- Delivered a production-ready platform trusted by UNESCO for ongoing global heritage site monitoring and international decision-making.

CONTACT

-  +61 434 569 700
-  will@auspatious.com
-  Melbourne, Australia
-  auspatious.com

SKILLS

- Full-Stack Geospatial Development
- Software Engineering and Lifecycle Management
- Cloud-Native Geospatial Architecture
- Spatial Data Engineering & Automation
- Cloud Infrastructure & DevOps
- Advanced Spatial Analysis & Visualisation
- Technical Leadership & Project Delivery

LANGUAGES

- Python
- JavaScript
- TypeScript
- SQL

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CAREER HISTORY

- **Senior Software Engineer**
Auspatious
- **Senior Geospatial Developer**
Mapular / Orbica
- **Geospatial Developer**
Orbica

EDUCATION

- **Postgraduate Diploma in GIS (Distinction)**
University of Canterbury
- **Bachelor of Science in Geography**
University of Canterbury

TRANSPARENC CARBON HUB

- Led architecture and development of a carbon monitoring platform providing near real-time spatial analysis for climate projects, managing a team of four developers from delivery through to handover.
- Built scalable Azure cloud infrastructure with Terraform and a geospatial analytics pipeline using Python, Airflow, PostGIS, COGs, and GeoParquet.
- Delivered a React/Next.js/Mapbox frontend with dynamic carbon data visualisation enabling clients to monitor and report on climate project performance.

BIODIVERSITY NET GAIN MAP CALCULATOR

- Built a national-scale web application supporting England's mandatory Biodiversity Net Gain scheme, enabling land professionals to assess and report biodiversity impact.
- Designed an on-demand cloud-native analysis pipeline using Python and PostGIS to generate bespoke biodiversity reports from user-submitted land parcels.
- Delivered automated PDF reports with maps, charts, and tables via email and in-app, with Stripe payment integration and secure user management across a React, Next.js, and Mapbox stack.

KIMODIS GROUNDWATER RESEARCH PLATFORM

- Delivered a scalable web-based analytics platform aggregating more than fifty years of groundwater and weather data across Germany for scientific research.
- Built a high-performance ETL pipeline using Airflow, Python, PostGIS, and GeoParquet, and served a fully authenticated data and tiling API via TiPG, FastAPI, and Auth0.
- Developed a secure, responsive web application using React, Next.js, Mapbox, and D3 for spatio-temporal visualisation across thousands of boreholes.

LEBENSWALD (LIVING FOREST) REFORESTATION PLATFORM

- Delivered a high-impact geospatial web application supporting orangutan habitat reforestation in Borneo, under tight time and budget constraints.
- Built a full-stack platform using React, PostGIS, and Azure with Mapbox-based interactive land sponsorship, donor display, and secure donation processing.

LORD HOWE ISLAND RODENT ERADICATION

- Supported a pest eradication program on World Heritage-listed Lord Howe Island through precision spatial tooling.
- Generated a bait placement grid engineered for complete rodent coverage and build custom Fulcrum spatial data collection forms for field teams logging inspections across extensive bait networks.
- Delivered tools that directly enabled a conservation outcome on one of Australia's most ecologically significant islands.